FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Congenital Malformations of the Genital Tract

Lecture | Week 5 · Bleeding, cancer & fertility | Gynaecology | Dr Kaduyu

Congenital genital tract anomalies arise when the paired Müllerian (paramesonephric) ducts fail to form, fuse or resorb their dividing septum, or when the vaginal plate fails to canalise. The results range from imperforate hymen and vaginal septa to MRKH syndrome (vaginal and uterine agenesis) and the septate, bicornuate, unicornuate and didelphys uteri. They present as primary amenorrhoea with cyclical pain, obstructed menstruation, dyspareunia, miscarriage, preterm birth or malpresentation. They are often associated with renal anomalies, because the Müllerian and Wolffian ducts develop side by side.

Must know before the ward round

1. Müllerian ducts form the tubes, uterus, cervix and upper vagina. The lower vagina comes from the urogenital sinus. The ovaries develop separately, so ovarian function and secondary sexual characteristics are normal in Müllerian anomalies.
2. Mechanisms: failure to develop (MRKH, unicornuate), failure of lateral fusion (didelphys, bicornuate), failure of septal resorption (septate), failure of vertical fusion or canalisation (transverse septum, cervical agenesis, imperforate hymen).
3. Primary amenorrhoea with normal breasts plus cyclical pain suggests an obstructed outflow (imperforate hymen or transverse septum). Without cyclical pain and with a short blind vagina, it suggests MRKH (46,XX, normal pubic hair) vs CAIS (46,XY, scant pubic hair, testes).
4. Imperforate hymen causes a bulging bluish membrane with haematocolpos, and sometimes urinary retention. Treat by cruciate or elliptical hymenotomy and drainage. Never needle-aspirate (causes pyocolpos).
5. Always scan the kidneys. Renal agenesis or ectopia accompanies about 30–40% of MRKH and unicornuate uteri. OHVIRA / Herlyn–Werner–Wunderlich = didelphys + obstructed hemivagina + ipsilateral renal agenesis.
6. Septate vs bicornuate depends on the external fundal contour (flat or minimally indented = septate; a cleft = bicornuate), shown on 3D ultrasound or MRI. Only the septate uterus is treated by hysteroscopic resection.
7. A non-communicating rudimentary horn pregnancy ruptures, often in the second trimester, with catastrophic haemorrhage. Excise a functional non-communicating horn.
8. ASRM 2021 classification has 9 categories: Müllerian agenesis, cervical agenesis, unicornuate, didelphys, bicornuate, septate, longitudinal vaginal septum, transverse vaginal septum, complex anomalies.
9. Primary amenorrhoea: investigate if no periods by about 15 with normal breasts (older texts 16) or by 13 with no breast development; always do a pregnancy test first.
10. Breasts? Pain? Hair? Breasts + cyclical pain = outflow obstruction; breasts, no pain, normal hair = MRKH (46,XX); breasts, no pain, scant hair = CAIS (46,XY); no breasts = Turner, Swyer or hypothalamic.
11. A normal hymenal ring above which the vagina ends blindly means the block is higher (transverse septum): get an MRI for level and thickness before surgery, and suppress menses as a bridge (norethisterone 5 mg 2–3 times daily).

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1. Definition

Congenital malformations of the female genital tract are developmental anomalies of the Müllerian (paramesonephric) duct system and the vaginal plate, producing abnormalities of the uterus, cervix, tubes and vagina (and hymen).

- The uterus, cervix, fallopian tubes and upper vagina are built from **two tubes (the Müllerian ducts) that must grow, meet in the midline, fuse, and then dissolve the wall between them**. The lower vagina must also **join up with, and open into, the urogenital sinus**.
- **Something going wrong at each step** produces a predictable anomaly:
  - **No ducts**: MRKH (no uterus, no upper vagina).
  - **One duct**: a unicornuate uterus.
  - **No fusion**: didelphys (two uteri, two cervices).
  - **Partial fusion**: bicornuate.
  - **No septal resorption**: septate.
  - **No canalisation or no vertical fusion**: transverse vaginal septum, imperforate hymen, cervical agenesis.
- The ovaries develop separately, so **these girls have normal ovaries, normal breasts and normal pubic hair**. That is the key to diagnosis in primary amenorrhoea.
- Because the **kidneys develop from the neighbouring Wolffian (mesonephric) system**, **renal anomalies are common**. Always scan the kidneys.
- **Never forget**: never needle-aspirate a haematocolpos, and never label a girl "pregnant" or "infertile" before examining her carefully and scanning her.

Key terms

Term	Meaning
Imperforate hymen	The hymenal membrane has no opening, so menstrual blood collects in the vagina (haematocolpos)
Haematocolpos / haematometra / haematosalpinx	Blood collected in the vagina / uterus / tubes
Mucocolpos	Mucus collected in the vagina in a newborn (stimulated by maternal oestrogen)
MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome	Congenital absence of the uterus and upper two-thirds of the vagina in a 46,XX female with normal ovaries. **Type I** is isolated. **Type II** has associated renal, skeletal or hearing anomalies (**MURCS**: Müllerian, Renal, Cervicothoracic Somite anomalies)
OHVIRA / Herlyn–Werner–Wunderlich (HWW)	**O**bstructed **H**emi**V**agina and **I**psilateral **R**enal **A**nomaly: uterus didelphys with one blocked hemivagina, and absent kidney on the same side

Term	Meaning
DSD	Differences (disorders) of sex development: a mismatch between chromosomal, gonadal and anatomical sex
Primary amenorrhoea	Investigate if **no periods by about 15 with normal breasts** (older texts say 16), or **by 13 with no breast development**; earlier if there is cyclical pain or a mass

2. Epidemiology

- Müllerian anomalies are found in about **5–6% of the general female population**, about **8% of infertile women**, about **13% of women with recurrent miscarriage**, and about **25% of women with both miscarriage and infertility**. Many are **asymptomatic**.
- The **septate uterus** is the commonest structural uterine anomaly in most series, followed by the bicornuate uterus.
- **Imperforate hymen**: about 1 in 1,000–2,000 girls. **Transverse vaginal septum**: about 1 in 30,000–80,000. **MRKH**: about **1 in 4,500–5,000** female births. It is the **second commonest cause of primary amenorrhoea** after gonadal dysgenesis (Turner syndrome).
- **Uganda context**:
  - There are few local prevalence data. Anomalies are usually found **incidentally at CS or ultrasound**, during work-up for **infertility or recurrent miscarriage**, or when a girl presents in **acute retention or with an abdominal mass**.
  - These girls often present late: a 14-year-old with **urinary retention and a pelvic mass** (haematocolpos) who is thought to be pregnant or to have a tumour, or a young wife who **cannot consummate her marriage or conceive** (MRKH), or a woman with **recurrent miscarriage or breech presentation**.
  - **Ultrasound** is widely available; **MRI** is available mainly in Kampala (Mulago and private centres) and is costly.
  - **Specialist reconstruction** (vaginoplasty, septum resection, horn excision) is done at national and regional referral and private hospitals.
  - **IVF** is available at a small number of private centres. **Gestational surrogacy** options for women with MRKH are limited and not clearly regulated.

> **UGANDA CONTEXT**
> Culturally, fertility and the ability to have intercourse are central to marriage (including bride-price expectations). A diagnosis of MRKH or of a DSD can bring **stigma, rejection and depression**. Counselling must be **private, sensitive and staged**, and involve the girl first and family members only with her consent. Watch for pressure towards early or inappropriate surgery.

3. Aetiology and Risk Factors

- **Mechanism-based causes** (section 4): failure of Müllerian duct **formation** (MRKH, unicornuate), **lateral fusion** (didelphys, bicornuate), **septal resorption** (septate, arcuate), **vertical fusion or canalisation** (transverse septum, cervical agenesis, imperforate hymen).
- **Associated Wolffian (mesonephric) insult**: renal agenesis or ectopia on the same side (unicornuate uterus, OHVIRA).
- **Genetic / familial**: most Müllerian anomalies are sporadic; rare familial cases. **CAIS** is X-linked (family history of amenorrhoea, infertility or "girls with hernias").
- **Drugs**: maternal diethylstilbestrol (DES), historically (AFS class VII).

- **DSD**: 46,XX virilisation from **CAH**; 46,XY DSD (CAIS, Swyer syndrome, 5-alpha reductase deficiency); 45,X Turner syndrome.

4. Pathophysiology

1. The indifferent stage (weeks 5–7)

- Both sexes have **two pairs of ducts**. The **mesonephric (Wolffian) ducts** drain the primitive kidney, and the **paramesonephric (Müllerian) ducts** form lateral to them from an invagination of the coelomic epithelium at about **week 6**.
- The gonad (the genital ridge, populated by germ cells migrating from the yolk sac) is still indifferent.

2. Sex determination: male vs female pathway

```
46,XY -> SRY -> testis
   Sertoli cells -> AMH -> Mullerian ducts REGRESS
   Leydig cells  -> testosterone -> Wolffian ducts -> epididymis, vas, seminal vesicle
                 -> DHT (5-alpha reductase) -> penis, scrotum, prostate
46,XX (no SRY) -> ovary
   no AMH -> Mullerian ducts PERSIST -> tubes, uterus, cervix, upper vagina
   no testosterone -> Wolffian ducts regress (remnants: Gartner duct cysts,
                      epoophoron, paroophoron)
   no DHT -> external genitalia stay female
```

Key concept: the female internal tract is the **default** when there is no AMH. The Wolffian ducts need **testosterone** to survive. External virilisation needs **DHT** acting through the **androgen receptor**.

3. Building the uterus: the Müllerian ducts

1. **Formation**: the ducts grow caudally beside the Wolffian ducts. If they do not form, the result is **agenesis (MRKH)**. If one fails, the result is a **unicornuate** uterus.
2. **Lateral fusion** (about weeks 8–12): the lower parts meet and fuse in the midline to form the **uterovaginal canal**. The unfused upper parts become the **fallopian tubes**.
   - Complete failure gives **didelphys**.
   - Partial failure at the fundus gives **bicornuate**.
3. **Septal resorption** (complete by about **week 20**): the midline wall between the fused ducts dissolves from **caudal to cranial**.
   - Failure gives a **septate** uterus (partial or complete).
   - Minimal residual gives an **arcuate** uterus.
4. **Vertical fusion and canalisation**: the fused Müllerian tip (the Müllerian tubercle) meets the **urogenital sinus**. That induces the **sinovaginal bulbs**, which form a solid **vaginal plate**. The plate then **canalises** (hollows out) by about 20 weeks.
   - Failure of the Müllerian and sinus parts to meet or canalise gives a **transverse vaginal septum** or **cervical agenesis**.
   - The **hymen** is the thin membrane left at the junction of the vaginal lumen and the urogenital sinus. If it does not perforate, the result is an **imperforate hymen**.

4. Why the kidneys are involved. The **Müllerian ducts form alongside, and depend on, the Wolffian ducts**, which also give rise to the **ureteric bud** (which induces the kidney). An insult to one side of the Wolffian system can therefore cause **both an absent kidney and a missing or obstructed Müllerian structure on the same side**. This explains the **unicornuate uterus with ipsilateral renal agenesis** and **OHVIRA**.

5. Why ovaries and hormones are normal. The ovaries come from the **genital ridge**, not the Müllerian ducts. So in MRKH, septa and fusion anomalies, the woman **ovulates, develops breasts and pubic hair normally, and has a normal 46,XX karyotype**. That is why these girls present with **amenorrhoea despite normal puberty**, and why **their eggs can be used in IVF**.

Mullerian ducts appear (week 6)
↓
Formation fails: MRKH or unicornuate | Lateral fusion (weeks 8-12)
↓
Fusion fails: didelphys or bicornuate | Septal resorption (by week 20)
↓
Resorption fails: septate or arcuate | Vertical fusion and canalisation with urogenital sinus
↓
Fails: transverse septum, cervical agenesis, imperforate hymen
↓
Menstrual blood trapped: haematocolpos, cyclical pain, retention
↓
Retrograde menstruation: endometriosis, tubal damage

Feature links

- Normal ovaries (genital ridge) → normal breasts, pubic hair and ovulation despite amenorrhoea.
- Outflow obstruction + working endometrium → blood trapped → **cyclical pain without bleeding**, haematocolpos/haematometra.
- Haematocolpos pressing on the bladder neck, rectum or sacral nerves → **urinary retention**, constipation, back pain.
- Blood behind a thin membrane at the introitus → **tense, bulging, bluish membrane**, more on Valsalva.
- No uterus (MRKH, CAIS) → amenorrhoea **without** cyclical pain; short blind vagina.
- Androgen receptor failure (CAIS) → testes make testosterone (male range) but no effect → **scant pubic and axillary hair**; testes aromatise to oestrogen → breasts; testes may present as **inguinal hernias**.
- One side drains, the other blocked (OHVIRA, functional non-communicating horn) → **regular periods with worsening unilateral pain** and a paravaginal mass.
- Poorly vascularised septum → poor implantation → **miscarriage** (septate).
- Small or divided cavity → **mid-trimester loss, preterm birth, cervical insufficiency, FGR, malpresentation**, retained placenta.
- Pregnancy in a thin-walled rudimentary horn → **rupture** (often second trimester) → haemorrhage.
- Long-standing obstruction → retrograde menstruation → **endometriosis**, tubal damage, infertility.
- Shared Wolffian origin → **ipsilateral renal agenesis or ectopia**.

5. Classification / Staging

ASRM Müllerian Anomalies Classification 2021 (MAC2021). The American Society for Reproductive Medicine replaced the 1988 AFS system with **nine descriptive categories**, illustrated with diagrams and designed to be easy to use:

#	Category	Key features
1	**Müllerian agenesis** (MRKH)	Absent or rudimentary uterus and upper vagina; may have rudimentary uterine "bulbs"

#	Category	Key features
2	**Cervical agenesis**	Absent or dysgenetic cervix, with or without the vagina; a functional uterus causes haematometra
3	**Unicornuate uterus**	One horn, with or without a rudimentary contralateral horn (communicating or not; with or without a cavity)
4	**Uterus didelphys**	Two separate uteri and two cervices, often with a longitudinal vaginal septum
5	**Bicornuate uterus**	An external fundal cleft; two cavities joined lower down; one cervix (or two)
6	**Septate uterus**	Normal or near-normal external contour; an internal septum (partial or complete)
7	**Longitudinal vaginal septum**	Obstructing or non-obstructing
8	**Transverse vaginal septum**	At any level; complete (obstructing) or with a small perforation
9	**Complex anomalies**	Combinations, e.g. OHVIRA, cervical duplication with septate uterus

> NOTE
> You will still see the **1988 AFS classes** (I agenesis or hypoplasia, II unicornuate, III didelphys, IV bicornuate, V septate, VI arcuate, VII DES-related) and the **ESHRE/ESGE 2013** system (U0–U6 for the uterus, C0–C4 for the cervix, V0–V4 for the vagina). Most classifications now regard the **arcuate** uterus (a mild fundal indentation of the cavity) as a **normal variant** with little clinical impact.

MRKH: type I (isolated) vs type II (with renal, skeletal or hearing anomalies; MURCS).

6. Clinical Features

History

Most of these patients are adolescent girls with **primary amenorrhoea**, sometimes with **cyclical pain, a pelvic mass or urinary retention**; a smaller group are women worked up for **recurrent miscarriage, preterm birth, malpresentation or infertility**, and a few are **newborns with ambiguous genitalia**.

Ask	Why it changes the plan
Age; when did breasts and pubic hair start; growth spurt	Breasts = working ovaries (oestrogen): the cause is **anatomical (outflow or Müllerian) or CAIS**. No breasts = **gonadal or hypothalamic** (Turner, Swyer, low weight, Kallmann) or constitutional delay
Cyclical (monthly) lower abdominal or back pain, but no bleeding	**Blood is made but trapped**: imperforate hymen, transverse septum, cervical agenesis, or a functional non-communicating horn
Urinary retention, frequency, constipation, pain on defecation, back pain	Mass effect of a haematocolpos on the bladder neck, rectum or sacral nerves
Periods regular but worsening one-sided pain and dysmenorrhoea, and a unilateral pelvic mass	**OHVIRA** or a functional non-communicating horn: an obstruction can coexist with normal periods (one side drains, the other is blocked)
Sexual activity; tampons; pain or failure at intercourse; dyspareunia; "a double vagina"	Short or blind vagina (MRKH, CAIS, septum); a longitudinal septum; decides whether a speculum is appropriate
Obstetric history: miscarriages (trimester), preterm births, repeated breech or transverse lies, retained placenta, cervical insufficiency	**Septate** (first-trimester and recurrent loss), **bicornuate/unicornuate** (mid-trimester loss, preterm birth, malpresentation)

Ask	Why it changes the plan
Infertility; heavy dysmenorrhoea	Septate uterus; **endometriosis** from retrograde menstruation through a long-standing obstruction
Kidney problems, UTIs, a single kidney; back or spine problems (scoliosis); hearing loss	Renal agenesis/ectopia; **MURCS** (type II MRKH)
Sense of smell, weight change, exercise, chronic illness	Kallmann syndrome, hypothalamic amenorrhoea
Family history of amenorrhoea, infertility or "girls with hernias"; maternal drugs (DES, historically)	Familial CAIS (X-linked); rare familial cases
Previous surgery: hernia repair (testes found?), "cutting" of the hymen, aspiration of a mass	CAIS; re-closure or pyocolpos
Neonate: ambiguous genitalia, vomiting or dehydration, an interlabial bulge	**CAH** (salt-wasting crisis), mucocolpos
Who knows, and what she wants to know	Staged, confidential disclosure; marriage pressure

Examination

Follow the order of Hutchison's Clinical Methods (24th edition). In an emergency (retention, sepsis, ruptured horn, salt-wasting baby) resuscitate first (section 9).

Introduction, consent and preparation

- Introduce yourself; **privacy**; an adolescent is examined with a **chaperone**, with a guardian's **consent** plus her own **assent**; explain every step.
- **Hand hygiene**.
- **Position and exposure**: supine for the abdomen; **lithotomy or frog-leg** for the genitalia; expose only what is needed.

General inspection and vital signs

- **Height and weight (BMI)**; plot growth.
- **Dysmorphic features**: **Turner syndrome** (short stature, webbed neck, widely spaced nipples, wide carrying angle); spine (**scoliosis**); hearing.
- **Vital signs** (pulse, BP, temperature) and **pain score**.

Hands, face and neck; chest

- Webbed neck, low hairline (Turner); widely spaced nipples.
- **Tanner staging**: breasts **and** pubic and axillary hair, separately. **Normal breasts with scant pubic and axillary hair = CAIS** until proven otherwise.

Abdomen and groins

- **Suprapubic mass** (haematocolpos or haematometra, or a distended bladder: re-examine after catheterising), tenderness.
- **Inguinal hernias or masses** in a "girl" (testes in CAIS).

Genitalia (gentle, chaperoned)

Look for	Interpretation
Clitoral size, labial fusion, position of the urethral opening	Virilisation: DSD (CAH, 5-alpha reductase deficiency)
Tense, bulging, bluish membrane that bulges more on Valsalva; no separate hymenal ring	**Imperforate hymen** with haematocolpos

Look for	Interpretation
Normal hymen with a **short blind vagina** behind it and **no bulge**	**Transverse septum** (mass above it on rectal exam), **MRKH** or **CAIS** (no mass)
Vaginal length, measured gently with a moistened **cotton-tipped swab** (**never force a speculum** in a girl who is not sexually active)	Short blind pouch = agenesis or septum
Speculum (sexually active only)	**Longitudinal septum**, **two cervices** (didelphys), a unilateral **bulge in the lateral vaginal wall** (obstructed hemivagina)

Rectal (rectoabdominal) examination

- Cystic, **bulging mass anteriorly** = **haematocolpos** (and its lower level: close to the introitus or high); **no uterus** palpable = MRKH or CAIS.

Completing the examination

- Thank her, cover her, hand hygiene; explain findings privately; summarise.

Red flags

RED FLAG

- **Acute urinary retention in a pubertal girl without periods**: haematocolpos until proven otherwise. Catheterise and examine the hymen.
- **Fever and pelvic pain with a known obstruction**: **pyocolpos or pyometra** (often after needle aspiration). Needs drainage and antibiotics.
- **First- or second-trimester pregnancy with abdominal pain and shock in a woman with a unicornuate uterus**: **rupture of a rudimentary horn pregnancy**. Resuscitate and do a laparotomy.
- **A newborn with ambiguous genitalia who is vomiting, dehydrated or collapsing at 1–3 weeks**: **salt-wasting CAH**. Check electrolytes and glucose, and give IV hydrocortisone.
- **A "girl" with inguinal hernias**: consider **CAIS**. Karyotype before the hernia repair.

Thinking like a clinician

1. **Primary amenorrhoea: ask two questions.**
   - "**Has she developed breasts?**" Yes means the ovaries are producing oestrogen, so look for an anatomical cause or CAIS.
   - "**Does she have monthly pain?**" Yes means blood is trapped, so look for an **outflow obstruction**. No means **there is no functional uterus** (MRKH) or no uterus at all (CAIS).
2. **MRKH or CAIS?** Both have breasts and a short blind vagina with no uterus.
   - **MRKH**: **46,XX**, **normal pubic and axillary hair**, ovaries, **female-range testosterone**.
   - **CAIS**: **46,XY**, **scant or absent pubic hair**, **testes** (inguinal or abdominal), **male-range testosterone**.
   - **The karyotype settles it.**
3. **Imperforate hymen or transverse septum?** An imperforate hymen **bulges at the introitus**, and there is **no hymenal ring visible separately**. A transverse septum sits **higher**: the hymen is normal, the vagina is short, and there is **no bulge at the introitus**. Getting this wrong is dangerous. A "hymenotomy" on a septum does not work and scars the vagina.
4. **Septate or bicornuate?** Look at the **outside of the fundus**. **Flat or minimally indented** means **septate** (treatable hysteroscopically). A **deep external cleft** means **bicornuate** (hysteroscopic resection would

perforate the uterus). An HSG shows only the cavity and **cannot distinguish them**. Use 3D ultrasound, MRI, or laparoscopy plus hysteroscopy.

5. **Regular periods do not exclude an obstruction.** In OHVIRA, one side drains normally and the other side fills, so she has increasing pain and a paravaginal mass.
6. **Always scan the kidneys.** Also think about the spine and hearing in MRKH.
7. **Always do a pregnancy test first**, however young she is and whatever she or her family says. Ask for it privately.
8. **Common student mistakes**:
   - calling a pelvic mass in a 13-year-old "pregnancy" or a "tumour" without looking at the hymen
   - needle aspiration of a haematocolpos
   - not checking pubic hair and inguinal regions (CAIS)
   - forgetting the renal scan

> EXAM FAVOURITE
> "Breasts? Pain? Hair?" Breasts + pain = obstruction. Breasts + no pain + normal hair = MRKH. Breasts + no pain + no hair = CAIS. No breasts = Turner, Swyer or hypothalamic.

7. Differential Diagnoses

Primary amenorrhoea with normal breast development

Differential	Distinguishing clinical feature	Distinguishing investigation
Imperforate hymen	Cyclical pain, a **bulging bluish membrane**, suprapubic mass, sometimes retention	Clinical; ultrasound showing haematocolpos
Transverse vaginal septum	Cyclical pain, normal hymen, **short blind vagina**, a mass felt on rectal exam	Ultrasound or **MRI** (level and thickness)
Cervical agenesis	Cyclical pain, haematometra, no cervix	MRI
MRKH	**No cyclical pain**, normal pubic hair, short blind vagina, no uterus	Ultrasound or MRI; **karyotype 46,XX**; renal US
CAIS	**Scant pubic and axillary hair**, inguinal masses, no uterus	**Karyotype 46,XY**, **testosterone in the male range**
Constitutional delay (breasts developing)	Family history of late menarche, otherwise normal	Watch; ultrasound shows a normal uterus
Pregnancy	Always consider (even in a "primary" presentation)	**Pregnancy test**

Primary amenorrhoea without breast development

Differential	Distinguishing clinical feature	Distinguishing investigation
Turner syndrome (45,X)	Short stature, webbed neck, streak ovaries, **uterus present**	High FSH; karyotype
Swyer syndrome (46,XY gonadal dysgenesis)	Tall, uterus present, streak gonads (high tumour risk)	High FSH; karyotype 46,XY
Hypothalamic / pituitary	Low weight, exercise, anosmia (Kallmann), chronic illness	Low FSH and LH; MRI of the pituitary

Pelvic mass in an adolescent

Differential	Distinguishing clinical feature	Distinguishing investigation
Haematocolpos / haematometra	Primary amenorrhoea, cyclical pain	Ultrasound
Pregnancy	Sexual activity (often denied)	Pregnancy test
Ovarian cyst or tumour (dermoid, germ cell tumour)	Adnexal mass, periods may be normal	Ultrasound, tumour markers
Full bladder	Retention	Mass disappears after catheterisation
Obstructed hemivagina	Regular periods, unilateral pain, paravaginal bulge	Ultrasound/MRI; renal US (ipsilateral agenesis)
Pelvic kidney	Asymptomatic	Renal ultrasound

8. Clinical Workup

The workup answers: **Is she pregnant? Is there an emergency (retention, infection, a ruptured horn, a salt-wasting baby)? Are the ovaries working (breasts)? Is blood trapped (cyclical pain), and where exactly is the block? Is there a uterus, and what is the karyotype? Are the kidneys normal?** Three rules: **pregnancy test first**; **always scan the kidneys**; **define the anatomy fully (MRI for anything above the hymen) before the first operation**, because the first operation is the best chance.

Bedside

"Available at" is the lowest level that usually offers the test Verify *availability compiled from the source notes, not an MoH list*.

Test	What you want	Why	Available at
Urine pregnancy test	hCG	Always first in amenorrhoea or a pelvic mass	All levels
Catheter volume (if retention)	Volume drained; does the mass shrink?	A mass that persists after emptying the bladder is haematocolpos/metra (or another mass)	All levels
Urine dipstick	Nitrites, leucocytes, blood	UTI from retention; renal anomaly clues	All levels
Point-of-care (transabdominal) ultrasound	Distended vagina or uterus full of echogenic fluid; is there a uterus? kidneys present?	Confirms haematocolpos quickly; **no uterus** points to MRKH/CAIS	HC IV and above with an operator
Hb / HemoCue	Anaemia	Before surgery; long-standing obstruction	HC III and above
Random blood sugar and electrolytes (neonate)	Hypoglycaemia, **low Na^+, high K^+**	**Salt-wasting CAH**	Glucose at all levels; electrolytes at hospitals
Vaginal length by swab	Blind short vagina	Agenesis vs septum	All levels

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
FBC	Before surgery; pyocolpos	Anaemia; raised WBC	Transfusion, antibiotics	HC IV and above
Urea, creatinine	Renal anomaly, single kidney, hydronephrosis	Raised creatinine	Protect the single kidney; avoid nephrotoxins	General hospital and above
Blood culture	Fever with suspected pyocolpos/pyometra	Organism	Antibiotic choice	Regional/national referral

Basic lab rows added to complete the workup; not in the source notes Verify *local practice*.

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Pelvic ultrasound (transabdominal; transperineal in adolescents; transvaginal if sexually active)	Every case, first line	Haematocolpos ± haematometra; uterus absent (MRKH/CAIS) or present; horns; ovaries (present in MRKH)	Diagnosis and urgency	Widely available (HC IV and above)
Renal ultrasound	**Always**	Renal agenesis, pelvic or horseshoe kidney, hydronephrosis	About **30–40%** association in MRKH and unicornuate uterus; protect a single kidney; OHVIRA side	HC IV and above
MRI pelvis	Transverse septum (level, thickness), cervical agenesis, obstructed hemivagina, rudimentary horns, before any reconstructive surgery	Exact anatomy	Surgical planning; the first operation is the best chance	**Kampala only** (Mulago and private), costly
3D ultrasound	Septate vs bicornuate	External fundal contour	Only the septate uterus is resected hysteroscopically	Specialist and private units
HSG	Infertility work-up	Cavity shape	**Cannot see the outer contour**, so cannot tell septate from bicornuate	Regional/national referral, private
Spine X-ray, audiology	MRKH	Vertebral anomalies, hearing loss	Type II MRKH (MURCS)	General hospital and above (audiology at referral)
Cervical length scans in pregnancy	Known bicornuate/unicornuate uterus or previous mid-trimester loss	Short cervix	Cerclage and/or vaginal progesterone	Referral and private units

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
Karyotype	No uterus (MRKH vs CAIS); no breasts (Turner, Swyer); ambiguous genitalia	46,XX / 46,XY / 45,X	Settles MRKH vs CAIS; gonadectomy decisions (Swyer at diagnosis, CAIS after puberty)	National referral, research or private laboratories; costly and slow
FSH, LH, oestradiol	Absent or poor breast development; suspected gonadal failure	**High FSH** = gonadal failure (Turner, Swyer); **low FSH/LH** = hypothalamic/pituitary	Directs to karyotype or pituitary work-up	Regional/national referral and private labs
Testosterone	No uterus with a blind vagina	**Male range** = CAIS; female range = MRKH	Supports the karyotype result	Regional/national referral and private labs
Prolactin, TSH	Primary amenorrhoea without an anatomical cause	Raised	Treat the endocrine cause	Regional/national referral and private labs
17-OH progesterone, electrolytes, glucose	Newborn with ambiguous genitalia	Raised 17-OHP; low Na^+, high K^+	**CAH**: hydrocortisone ± fludrocortisone; paediatric endocrinology	17-OHP mainly at national referral/private labs
Hysteroscopy ± laparoscopy	Recurrent loss or infertility with a suspected septum; endometriosis	Septum, cavity, fundal contour, endometriosis	Diagnosis and treatment in one sitting	Referral and private centres

Putting it together

Model one-line summary: "A 14-year-old with **primary amenorrhoea, normal breasts and pubic hair (Tanner 4)**, 4 months of **cyclical pain**, **acute urinary retention**, a 20-week suprapubic mass and a **bulging bluish membrane at the introitus**; pregnancy test negative, ultrasound showing a **haematocolpos**, both kidneys normal: imperforate hymen."

Problem list

1. Primary amenorrhoea with outflow obstruction (imperforate hymen).
2. Haematocolpos with mild haematometra.
3. Acute urinary retention (catheterised).
4. Anxiety and family fear of pregnancy or tumour.

Working diagnosis and plan: imperforate hymen → catheter, analgesia, **cruciate or elliptical hymenotomy under anaesthesia with drainage** (no needle aspiration), follow-up for normal menses and stenosis.

Model case presentation (ward round)

> "N is a 14-year-old girl from Mukono brought with **acute urinary retention** and 4 months of **cyclical lower abdominal pain**. She has **not started her periods** despite **normal breast development (Tanner 4) and pubic hair**. On examination she has a **tender suprapubic mass** (about 20 weeks' size) and a **tense, bulging, bluish membrane at the introitus**. The urine pregnancy test is negative. Ultrasound shows a **large haematocolpos with mild haematometra**, and both kidneys are normal. **Summary: primary amenorrhoea due to imperforate hymen with haematocolpos causing urinary retention.** Plan: catheterise, analgesia, and **hymenotomy (cruciate incision) under anaesthesia with drainage**, with no needle aspiration, then follow up for normal menses."

Worked example 1: primary amenorrhoea with cyclical pain

Two 15-year-olds, both with normal breasts and pubic hair, monthly pain and no periods. Both pregnancy tests are negative.

	Girl A	Girl B
Introitus	**Bulging bluish membrane**, bulges more on Valsalva	**Normal hymenal ring**, pink, no bulge
Swab	Cannot pass	Enters a **blind vagina about 3 cm** long
Rectal exam	Large tense cystic mass low in the pelvis	Tense mass **higher up**
Ultrasound	Haematocolpos reaching the introitus	Haematocolpos above a short lower vagina
Kidneys	Normal	Normal
Diagnosis	**Imperforate hymen**	**Transverse vaginal septum** (mid-vagina)
Next step	Hymenotomy with drainage	**MRI** for level and thickness; **menstrual suppression** (norethisterone 5 mg 2–3 times daily) as a bridge; **septum resection** with end-to-end anastomosis by an experienced surgeon, then dilators

A **third pattern**: normal breasts, cyclical pain, a **normal-length vagina ending without a cervix**, and haematometra on scan = **cervical agenesis**: MRI and specialist referral.

> **RED FLAG**
> Doing a "hymenotomy" on Girl B would not reach the blood and would scar the lower vagina. **Look before you cut**: a normal hymenal ring means the block is higher.

Worked example 2: primary amenorrhoea without pain

Two 17-year-olds with normal breasts, no cyclical pain and a **short blind vagina**. Ultrasound in both: **no uterus**.

	Girl C	Girl D
Pubic and axillary hair	**Normal**	**Scant**
Inguinal regions	Normal	**Bilateral inguinal swellings** (previous "hernia repair" as a child)
Ovaries on scan	**Present**	Not seen; gonads in the groins
Testosterone	Female range	**Male range**
Karyotype	**46,XX**	**46,XY**
Kidneys	**Right renal agenesis**	Normal
Diagnosis	**MRKH** (type II if renal/spinal anomalies)	**CAIS**

	Girl C	Girl D
Plan	Staged counselling; **vaginal dilators first line**; spine and hearing check; no HRT needed; motherhood via IVF with her own eggs and a gestational carrier, or adoption	Staged, sensitive disclosure; dilators; **gonadectomy usually after puberty**, then oestrogen replacement; MDT

9. Management

Immediate / emergency

First 5 minutes

1. **A, B, C: vital signs** (pulse, BP, temperature) and **pain score**. Give analgesia early.
2. **Urine pregnancy test** in every girl or woman with amenorrhoea or a pelvic mass, however young and whatever she or her family says. Ask for it privately.
3. **Acute urinary retention** in a pubertal girl without periods: **catheterise** (gently; the urethra is stretched over the mass) and record the volume. Then **look at the hymen**: haematocolpos until proven otherwise.
4. **Fever with pelvic pain** and a known or suspected obstruction (especially after someone has "tapped" the mass): **pyocolpos or pyometra**. Blood culture if available, **IV ceftriaxone 1 g daily + metronidazole 500 mg 8-hourly**, and plan surgical drainage.
5. **Pregnant with a known unicornuate uterus, abdominal pain and shock**: **ruptured rudimentary horn pregnancy**. Two large-bore cannulae, cross-match, fluids and **laparotomy**.
6. **Newborn with ambiguous genitalia who is vomiting, dehydrated or collapsed** (often at 1–3 weeks): **salt-wasting CAH**. Check glucose and electrolytes, send 17-OH progesterone before treatment if possible, give **IV saline, glucose and hydrocortisone** per the paediatric protocol, and call the paediatrician. **Do not assign sex at the bedside.**
7. **Privacy and consent**: an adolescent is examined with a chaperone, and a guardian's consent plus her own assent; explain every step.

Definitive

A. General principles

1. **Diagnose the anatomy precisely before any surgery** (ultrasound, and MRI if complex). The first operation is the best chance.
2. **Relieve obstruction promptly**: pain, retention, and the risk of endometriosis, infection and tubal damage.
3. **Scan the kidneys** and assess other associated anomalies.
4. **Refer complex cases** to a paediatric or adolescent gynaecology or reconstructive centre (a national or regional referral hospital).
5. **Psychological support and staged, confidential counselling** for the girl and, with her consent, her family.
6. **Temporary menstrual suppression** can "buy time" before definitive surgery for an obstruction.

B. Imperforate hymen: step by step

1. **Confirm the diagnosis**: a bulging bluish membrane, and ultrasound showing haematocolpos. **Catheterise if she is in retention.**
2. **Do not needle-aspirate**. It drains incompletely, and the blood becomes infected (**pyocolpos**).
3. **Hymenotomy under anaesthesia** (GA or spinal, or ketamine per local practice), in **lithotomy**:
   - Make a **cruciate (X-shaped) or elliptical incision** in the membrane.
   - **Drain the thick, dark "chocolate" blood.** Do not probe or instrument the uterus.
   - **Excise the four quadrant flaps** back to the hymenal ring.

- **Suture the vaginal mucosa to the hymenal ring edges** with fine absorbable sutures (e.g. 3-0 or 4-0 polyglactin) to prevent re-closure.

4. **Afterwards**:
   - analgesia
   - sitz baths
   - **a single prophylactic antibiotic dose** is commonly given
   - check she can void
   - **follow up** to confirm normal menses and to check for stenosis or re-closure, and for endometriosis symptoms if the obstruction was long-standing
5. **Timing in children**:
   - An asymptomatic imperforate hymen found in infancy is ideally repaired **after oestrogenisation (around early puberty) but before menarche**.
   - A **symptomatic neonatal mucocolpos** (retention, mass effect) is drained at diagnosis.

C. Transverse vaginal septum

1. Define the **level (lower, mid or upper vagina)** and **thickness** by **MRI**. A thin, low septum is easier.
2. **Menstrual suppression** while waiting (see the drug table).
3. **Excision of the septum with end-to-end anastomosis** of the upper and lower vaginal mucosa, by an experienced surgeon. A thick septum may need **pre-operative dilator therapy**, grafts or flaps.
4. **Post-operatively**: **vaginal moulds or dilators** to prevent **stenosis**, which is the main complication.
5. A small perforated septum in pregnancy may cause **dystocia**. It can be incised at delivery or deliver by CS.

D. Longitudinal vaginal septum and OHVIRA

- **Non-obstructing longitudinal septum**: excise it only if it is symptomatic (dyspareunia, difficulty with tampons) or it obstructs labour. It can often be divided in the second stage with ligatures if it tightens across the presenting part.
- **OHVIRA (obstructed hemivagina)**:
  - **resect the obstructing vaginal septum** to drain the blocked side and create one vagina
  - **do not remove the uterus**
  - the renal agenesis needs no treatment except **protecting the single kidney**: avoid nephrotoxins, and check BP and renal function periodically
  - pregnancy outcomes after septum resection are generally good

E. MRKH syndrome

1. **Disclosure and counselling first**:
   - Do it privately, with the girl, in stages.
   - Emphasise that **she is a woman, with normal ovaries and hormones**.
   - A **functional vagina can be created**.
   - Motherhood is possible through **IVF with her own eggs and a gestational carrier (surrogate)** or adoption. Uterine transplantation is experimental.
   - Offer **peer support** and psychology.
2. **Creating a vagina**:
   - **Timing**: when **she** is emotionally mature and motivated, usually late adolescence. **Never rush it** because of marriage pressure.
   - **First line: progressive vaginal dilator therapy (Frank method)**. Graduated dilators are pressed against the vaginal dimple for about 20–30 minutes once or twice daily over months. This is effective in most motivated women, and has no surgical risk. ACOG recommends it first line.

- **Surgery** if dilation fails or she chooses it:
  - **McIndoe** (split-thickness skin graft over a mould)
  - **Vecchietti** (traction device, laparoscopic)
  - **Davydov** (peritoneal lining)
  - **bowel vaginoplasty** (sigmoid)
  - all need **ongoing dilation or intercourse** to maintain the vagina

3. **Associated anomalies**: renal ultrasound, spine and hearing assessment.
4. **No HRT is needed** (the ovaries work). If rudimentary uterine horns contain functional endometrium and cause pain, laparoscopic excision may be needed.

F. Uterine anomalies

Anomaly	Main problems	Management
Septate	**Highest miscarriage risk** among the anomalies (the septum is poorly vascularised, so implantation is poor), preterm birth, malpresentation, infertility	**Hysteroscopic septum resection (metroplasty)** is offered to women with **recurrent pregnancy loss** (or after a poor outcome) after counselling. Recent trials have questioned its benefit in unselected women, so do not operate on an incidental septum in asymptomatic women
Bicornuate	Second-trimester loss, **preterm labour**, **cervical insufficiency**, malpresentation	Usually **no surgery**. **Serial cervical length** scans in pregnancy, with cerclage if the cervix shortens. Abdominal (Strassman) metroplasty only rarely, in selected recurrent loss
Unicornuate	Miscarriage, preterm birth, **FGR**, malpresentation. **Rudimentary horn pregnancy with rupture**	**Laparoscopic excision of a functional non-communicating rudimentary horn** (with the ipsilateral tube), to prevent horn pregnancy, haematometra and endometriosis. Monitor the cervix and fetal growth in pregnancy. **Renal scan**
Didelphys	Generally the **best outcomes** of the fusion defects; preterm birth, malpresentation; a **longitudinal septum**; OHVIRA	Usually none. Excise the vaginal septum if symptomatic. Metroplasty is not recommended
Arcuate	Usually none	No treatment
Cervical agenesis	Haematometra, endometriosis, infertility	Specialist management: **menstrual suppression**, then **uterovaginal anastomosis** in expert centres. Historically hysterectomy

G. Pregnancy in a woman with a known uterine anomaly

1. **Early ultrasound** to confirm the site (exclude a rudimentary horn pregnancy, and check which horn).
2. **Serial cervical length** (from 16–24 weeks) in bicornuate or unicornuate uterus, or with a previous mid-trimester loss. **Cerclage and/or vaginal progesterone** as indicated.
3. **Growth scans** (FGR, especially with a unicornuate uterus).
4. **Check the presentation at 36 weeks**. **ECV is relatively contraindicated** with a major uterine anomaly.
5. **Delivery**: vaginal birth is possible with a cephalic presentation. Anticipate **retained placenta and PPH**. CS rates are higher. **Document the anomaly** in the operation note for future care.

H. DSD: an overview for the student

- **Newborn with ambiguous genitalia**:
  - Do **not** assign sex at the bedside.
  - Use neutral language ("your baby").

    - **Exclude CAH urgently** (17-OH progesterone, electrolytes, glucose): it is the **commonest cause of ambiguous genitalia in a 46,XX baby**, and salt-wasting can kill in the second or third week.
    - Refer to a **multidisciplinary team** (paediatric endocrinology, surgery, genetics, psychology).
- **46,XY DSD**:
    - **CAIS**: female phenotype, blind vagina, testes. **Gonadectomy**, usually **after puberty** (the testes provide oestrogen by aromatisation for natural breast development, and the tumour risk before adulthood is low), then **oestrogen replacement**. Disclosure is sensitive.
    - **Swyer syndrome**: streak gonads with a **high risk of gonadoblastoma**, so **gonadectomy at diagnosis**, then HRT. The uterus is present, so pregnancy is possible with donor eggs.
    - **5-alpha reductase deficiency**: virilises at puberty.
- **Turner syndrome (45,X)**: streak ovaries, short stature, cardiac (coarctation, bicuspid aortic valve) and renal anomalies. Growth hormone, oestrogen for puberty, and cardiac follow-up.

Drug table

Drug	Dose (adult; child per kg where stated)	Route	Frequency	Duration	Key cautions
Norethisterone (menstrual suppression while awaiting surgery)	5 mg	PO	2–3 times daily, continuously	Until definitive surgery (short term)	Bloating, spotting; a bridge, not a cure
Combined oral contraceptive (continuous, skipping pill-free weeks)	1 tablet (e.g. ethinylestradiol 30 µg + levonorgestrel 150 µg)	PO	Once daily, continuously	Until surgery	Alternative suppression; check UKMEC/WHO MEC eligibility
GnRH agonist (e.g. goserelin)	3.6 mg	SC	Every 28 days	Specialist; add-back therapy if more than 6 months	Specialist use, for cervical agenesis or complex obstruction; hypo-oestrogenic effects
Ibuprofen	400 mg	PO	8-hourly with food	As needed	Cyclical pain / dysmenorrhoea; avoid in renal impairment (single kidney)
Paracetamol	1 g (15 mg/kg in children)	PO / IV	6-hourly	As needed	Max 4 g/day (60 mg/kg/day in children)
Ceftriaxone + metronidazole (pyocolpos or pyometra)	1 g + 500 mg	IV	Ceftriaxone once daily; metronidazole 8-hourly	Switch to oral to complete about 7 days	Plus **surgical drainage**
Prophylactic antibiotic (hymenotomy)	Single dose per unit protocol	IV	Once	Single dose	Commonly given
Hydrocortisone (neonatal salt-wasting CAH crisis)	25 mg stat, then 50–100 mg/m²/day	IV	Stat dose, then the daily dose split 6-hourly per the paediatric protocol	Until stable, then maintenance per paediatric endocrinology	**Paediatric emergency**; plus IV saline and glucose; specialist guidance

DRUG DOSING

Menstrual suppression is a **bridge**, not a cure. It relieves cyclical pain and stops further accumulation while you arrange imaging and definitive surgery. It does **not** drain an existing haematocolpos.

Referral criteria

Referral triggers compiled from the source notes and UCG levels of care Verify *not an MoH referral list*.

Level	What to do	Refer up when
HC III	Pregnancy test, catheterise retention, analgesia, look at the hymen	Any suspected anomaly, pelvic mass, primary amenorrhoea → HC IV / general hospital
HC IV / General hospital	Pelvic and renal ultrasound, hymenotomy for a clear imperforate hymen, drainage and antibiotics for pyocolpos, laparotomy for ruptured horn, start menstrual suppression, resuscitation of a salt-wasting neonate	Normal hymen with a blind vagina (septum), no uterus (MRKH vs CAIS), cervical agenesis, OHVIRA, rudimentary horn, ambiguous genitalia → RRH / NRH
Regional referral hospital	Specialist gynaecology, hysteroscopy/laparoscopy where available, cervical length surveillance, karyotype sampling	MRI, karyotype/hormones, complex reconstruction (septum resection, vaginoplasty, uterovaginal anastomosis), DSD MDT → NRH / private specialist centres
National referral (Mulago) and private centres	MRI, karyotype, 17-OHP, paediatric endocrinology, reconstructive surgery, IVF (private)	n/a

Monitoring and follow-up

- **After hymenotomy**: confirm normal menses; check for stenosis or re-closure; ask about endometriosis symptoms if the obstruction was long-standing.
- **After septum resection or vaginoplasty**: vaginal moulds or dilators; review for stenosis.
- **MRKH on dilator therapy**: regular review of progress and psychological support.
- **Single kidney**: BP and renal function periodically; avoid nephrotoxins.
- **Pregnancy with a uterine anomaly**: early scan, serial cervical length from 16–24 weeks, growth scans, presentation at 36 weeks.
- **CAIS after gonadectomy**: oestrogen replacement; **Turner**: cardiac follow-up.

10. Complications

Of the anomaly (disease-related)	Of treatment (treatment-related)
Haematocolpos causing **retention**, **hydronephrosis**, constipation	**Pyocolpos** after needle aspiration or incomplete drainage
Endometriosis and tubal damage from retrograde menstruation (long obstruction) causing infertility	**Vaginal stenosis** after septum resection or vaginoplasty (without dilator use)
Pyometra or pyocolpos	Re-obstruction or re-closure after hymenotomy
Miscarriage, preterm birth, FGR, malpresentation, retained placenta, PPH	**Uterine perforation** at hysteroscopic septum resection; uterine rupture in a later pregnancy
Rudimentary horn pregnancy with rupture	Bowel or bladder injury (vaginoplasty)
Infertility	Psychological harm from poorly handled disclosure

Of the anomaly (disease-related)	Of treatment (treatment-related)
Psychosexual distress, marital breakdown, stigma	Gonadal tumour if gonadectomy is delayed in Swyer syndrome
Salt-wasting crisis (CAH)	Hysteroscopic resection of a bicornuate uterus: perforation

11. Prognosis

- **Imperforate hymen**: excellent after a correct hymenotomy, provided no long-standing endometriosis or tubal damage.
- **Transverse septum**: stenosis is the main problem; thick and high septa do worse; the first operation is the best chance.
- **MRKH**: a functional vagina is achieved by dilators in most motivated women; normal ovarian function; motherhood possible via IVF with her own eggs and a gestational carrier, or adoption.
- **Uterine anomalies**: **didelphys** has the best outcomes of the fusion defects; **septate** has the highest miscarriage risk; bicornuate and unicornuate carry mid-trimester loss, preterm birth, FGR and malpresentation. **OHVIRA**: pregnancy outcomes after septum resection are generally good.
- **Ugandan outcome data are not available.**

12. Prevention

- Most anomalies cannot be prevented (developmental, mostly sporadic); historically, avoid DES in pregnancy.
- **Prevent the complications**: early recognition of primary amenorrhoea with cyclical pain; **never needle-aspirate**; define anatomy before the first operation; dilators after reconstruction; excise a functional non-communicating horn before pregnancy; serial cervical length in pregnancy.
- **Prevent harm from DSD**: do not assign sex at the bedside; screen for CAH in ambiguous genitalia; karyotype a "girl" with inguinal hernias before hernia repair; gonadectomy at diagnosis in Swyer.
- **Prevent psychological harm**: private, staged disclosure; resist marriage pressure for early surgery.

13. Special Populations

- **Newborns**: mucocolpos (drain if symptomatic); ambiguous genitalia = salt-wasting CAH until excluded; do not assign sex at the bedside.
- **Children**: an asymptomatic imperforate hymen is repaired after oestrogenisation but before menarche; never force a speculum; measure vaginal length with a swab.
- **Adolescents**: guardian consent plus her assent; privacy; staged disclosure; dilator therapy only when she is ready.
- **Pregnant women with a known anomaly**: early scan for horn pregnancy, cervical surveillance, growth scans, ECV relatively contraindicated, anticipate retained placenta and PPH.
- **Women with a single kidney**: avoid nephrotoxins (NSAIDs with care); periodic BP and renal function.

14. Key Points

High-yield points

- **Form, fuse, dissolve, open**: each failure gives a predictable anomaly.
- **Ovaries are normal** in Müllerian anomalies: normal breasts and pubic hair.

- **Breasts? Pain? Hair?** sorts primary amenorrhoea.
- **Pregnancy test first**, always.
- **Imperforate hymen**: bulging bluish membrane; **hymenotomy, never needle aspiration**.
- **Normal hymenal ring + blind vagina** = block is higher: MRI before surgery; suppress menses as a bridge.
- **MRKH vs CAIS**: hair, gonads, testosterone, **karyotype**.
- **Always scan the kidneys** (30–40% association); OHVIRA = didelphys + obstructed hemivagina + ipsilateral renal agenesis.
- **Septate vs bicornuate**: external fundal contour (3D US/MRI); only septate is resected; HSG cannot tell.
- **Rudimentary horn pregnancy ruptures**: excise a functional non-communicating horn.

Numbers & doses

Item	Value
Müllerian anomalies in the general female population	About **5–6%** (about 8% infertile; about 13% recurrent miscarriage; about 25% miscarriage plus infertility)
Imperforate hymen	About **1 in 1,000–2,000** girls
Transverse vaginal septum	About **1 in 30,000–80,000**
MRKH	About **1 in 4,500–5,000** female births; **second commonest cause of primary amenorrhoea** after gonadal dysgenesis
Renal anomalies with MRKH or unicornuate uterus	About **30–40%**
Embryology	Müllerian ducts appear about **week 6**; lateral fusion about **weeks 8–12**; septal resorption and canalisation complete by about **week 20**
Investigate primary amenorrhoea	**No periods by about 15 with normal breasts** (older texts 16), or **by 13 with no breast development**; earlier with cyclical pain or a mass
ASRM MAC2021	**9 categories**
Hymenotomy sutures	Fine absorbable, e.g. **3-0 or 4-0 polyglactin**
Norethisterone (menstrual suppression)	**5 mg orally 2–3 times daily**, continuously
Goserelin (specialist)	**3.6 mg SC every 28 days**; add-back if more than 6 months
Pyocolpos/pyometra	**Ceftriaxone 1 g IV daily + metronidazole 500 mg IV 8-hourly** + drainage; about 7 days in total
Neonatal salt-wasting CAH	**Hydrocortisone 25 mg IV stat**, then 50–100 mg/m²/day in divided doses + IV saline and glucose; crisis typically at **1–3 weeks**
Vaginal dilator therapy (MRKH)	About **20–30 min, once or twice daily**, over months
Cervical length surveillance (bicornuate/unicornuate)	From about **16–24 weeks**
Presentation check with a uterine anomaly	**36 weeks** (ECV relatively contraindicated)

Exam traps

- **Needle-aspirating a haematocolpos.** Never: incomplete drainage and **pyocolpos**. Do a formal hymenotomy.
- **A "hymenotomy" for a transverse septum.** A normal hymenal ring means the block is higher: MRI and a specialist septum resection.

- **Labelling a 13-year-old with a pelvic mass as pregnant (or as having a tumour)** without doing a pregnancy test and looking at the hymen.
- **Using HSG to tell septate from bicornuate.** HSG shows only the cavity; you need the **external contour** (3D ultrasound, MRI or laparoscopy).
- **Hysteroscopic resection of a bicornuate uterus.** Perforation; only the **septate** uterus is resected, and only after counselling (recurrent loss).
- **"Regular periods exclude an obstruction."** Not in **OHVIRA** or a functional non-communicating horn.
- **Forgetting the kidneys** (and spine and hearing in MRKH) or the **pubic hair and groins** (CAIS).
- **Starting HRT for MRKH.** Not needed: the ovaries work. CAIS needs oestrogen only after gonadectomy.

Questions seniors ask

Viva questions

1. **What do the Müllerian ducts form?** The fallopian tubes, uterus, cervix and upper vagina. The lower vagina comes from the urogenital sinus.
2. **Why do girls with MRKH have normal breasts?** The ovaries come from the genital ridge, not the Müllerian ducts, so they produce oestrogen normally.
3. **How do you distinguish MRKH from CAIS?** MRKH: 46,XX, normal pubic hair, female testosterone. CAIS: 46,XY, scant pubic hair, testes, male testosterone.
4. **Why scan the kidneys?** The Müllerian and Wolffian ducts develop together, so renal agenesis or ectopia is common (about 30–40% in MRKH or unicornuate uterus).
5. **What is OHVIRA?** Didelphys, an obstructed hemivagina, and ipsilateral renal agenesis.
6. **How do you tell septate from bicornuate?** By the external fundal contour on 3D ultrasound or MRI (or laparoscopy). HSG cannot distinguish them.
7. **Which anomaly has the worst miscarriage rate?** Septate.
8. **What is the danger of a rudimentary horn?** A horn pregnancy ruptures (often in the second trimester), causing massive haemorrhage. Excise it.
9. **What is first-line treatment for vaginal agenesis?** Vaginal dilator therapy.

On the ward round

- **How do you work up primary amenorrhoea with cyclical pain?** Pregnancy test; Tanner staging; inspect the introitus (bulging membrane vs normal hymen), swab the vaginal length, rectal exam; pelvic and **renal** ultrasound; MRI if the block is above the hymen.
- **How do you tell MRKH from CAIS?** Pubic and axillary hair (normal vs scant), gonads (ovaries vs inguinal/abdominal testes), testosterone (female vs male range) and the **karyotype** (46,XX vs 46,XY).
- **Why does a girl with MRKH have normal breasts?** The ovaries come from the genital ridge, not the Müllerian ducts, so oestrogen production is normal.
- **What is the first-line treatment for vaginal agenesis?** Progressive **dilator therapy** when she is ready; surgery (McIndoe, Vecchietti, Davydov, bowel) if dilation fails or she chooses it.
- **What is OHVIRA and how is it treated?** Didelphys + obstructed hemivagina + ipsilateral renal agenesis; resect the obstructing vaginal septum, keep the uterus, protect the single kidney.
- **What is the danger of a non-communicating rudimentary horn?** Horn pregnancy ruptures (often in the second trimester) with massive haemorrhage; excise a functional horn with its tube.
- **A newborn has ambiguous genitalia. What do you do first?** Do not assign sex; check glucose and electrolytes; send 17-OHP and karyotype; watch for salt-wasting (hydrocortisone, saline); refer to the MDT.

Mnemonics

- **"Form, Fuse, Dissolve, Open"**: the 4 steps.

    - **Form** fails: agenesis / unicornuate.
    - **Fuse** fails: didelphys / bicornuate.
    - **Dissolve** fails: septate.
    - **Open** fails (canalise / vertical fusion): transverse septum, cervical agenesis, imperforate hymen.
- **Primary amenorrhoea: "Breasts? Pain? Hair?"**
    - Breasts plus pain means **obstruction**.
    - Breasts, no pain, and hair means **MRKH**.
    - Breasts, no pain, and **no hair** means **CAIS**.
    - No breasts means **Turner or Swyer or hypothalamic**.
- **OHVIRA spells it out**: **O**bstructed **H**emi**V**agina + **I**psilateral **R**enal **A**genesis.
- **MURCS**: **MÜ**llerian aplasia, **R**enal aplasia, **C**ervicothoracic **S**omite dysplasia.
- **"Septate = Surgery (hysteroscopic); Bicornuate = Be patient (watch the cervix)."**

OSCE tips

- In an **amenorrhoea history** station, ask about breasts, pubic hair, cyclical pain, urinary symptoms, growth, sense of smell, weight and exercise, family history, and sexual activity (and always consider pregnancy).
- In an **MRKH counselling** station: a private setting, check what she knows, give a **warning shot**, then give information in stages. Emphasise normal ovaries and womanhood, and the options for a vagina and for motherhood. Offer support, and ask who she wants involved.
- In an **examination** station: chaperone and assent, Tanner staging (breasts and hair separately), groins, inspect the introitus with and without Valsalva, never force a speculum, and state that you would do a rectal examination and a renal scan.

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SELF-TEST

Flashcards

25 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	What do the Müllerian (paramesonephric) ducts form?	The **fallopian tubes, uterus, cervix and upper vagina**. The lower vagina and hymen come from the **urogenital sinus**. The ovaries come from the genital ridge.
2	Why do Müllerian ducts persist in a female fetus?	There is **no AMH** (anti-Müllerian hormone from Sertoli cells), so the ducts persist by default. The Wolffian ducts regress because there is **no testosterone**.
3	Match each developmental failure to its anomaly.	**Formation** fails: agenesis (MRKH) or unicornuate. **Lateral fusion** fails: didelphys or bicornuate. **Septal resorption** fails: septate. **Vertical fusion or canalisation** fails: transverse septum, cervical agenesis. **Hymen** does not perforate: imperforate hymen.
4	ASRM 2021 (MAC2021) categories?	Nine: **Müllerian agenesis, cervical agenesis, unicornuate, didelphys, bicornuate, septate, longitudinal vaginal septum, transverse vaginal septum, complex anomalies.**
5	Why are renal anomalies common with Müllerian anomalies?	The Müllerian ducts develop alongside, and depend on, the **Wolffian (mesonephric) ducts**, which also give rise to the ureteric bud. So there is ipsilateral **renal agenesis or ectopia**, especially in MRKH and unicornuate uterus (about 30–40%).
6	Classic presentation of imperforate hymen?	A pubertal girl with **primary amenorrhoea**, normal breasts, **cyclical pain**, a suprapubic mass, sometimes **urinary retention**, and a **bulging bluish membrane** at the introitus.
7	Treatment of imperforate hymen, and what to avoid?	A **cruciate or elliptical hymenotomy** under anaesthesia, drain the old blood, excise the flaps, and suture the mucosa to the ring. **Never needle-aspirate** (it causes pyocolpos and incomplete drainage).
8	Imperforate hymen vs transverse vaginal septum?	The hymen **bulges at the introitus**. A septum lies **higher**, with a normal hymen and a **short blind vagina** and no bulge at the introitus. MRI defines the level and thickness of a septum. Excision needs a specialist and dilators afterwards.
9	What is MRKH syndrome?	**46,XX**, congenital absence of the **uterus and upper vagina**, **normal ovaries**, normal breasts and pubic hair. About 1 in 4,500–5,000 female births. Type I is isolated; type II (**MURCS**) has renal, skeletal and hearing anomalies.
10	MRKH vs CAIS?	**MRKH**: 46,XX, normal pubic and axillary hair, ovaries, female testosterone. **CAIS**: 46,XY, **scant pubic and axillary hair**, testes (inguinal or abdominal), **male-range testosterone**. Both have breasts and a blind vagina with no uterus.
11	First-line treatment for vaginal agenesis (MRKH)?	**Progressive vaginal dilator therapy** (Frank method), when she is mature and motivated. Surgery (McIndoe, Vecchietti, Davydov, bowel) if it fails.
12	What is OHVIRA (Herlyn–Werner–Wunderlich)?	**O**bstructed **H**emi**V**agina and **I**psilateral **R**enal **A**genesis, with a **uterus didelphys**. Regular periods but increasing dysmenorrhoea and a paravaginal mass. Treated by **resecting the vaginal septum**.
13	How do you distinguish septate from bicornuate uterus?	By the **external fundal contour**: flat or minimally indented means septate; a deep cleft means bicornuate. Use **3D US or MRI** (or laparoscopy plus hysteroscopy). HSG cannot distinguish them.

#	Question	Answer
14	Which uterine anomaly has the highest miscarriage risk and how is it treated?	The **septate uterus** (the poorly vascularised septum impairs implantation). Offer **hysteroscopic septum resection** to selected women with recurrent loss, after counselling (the benefit is debated in unselected women).
15	Danger of a non-communicating rudimentary horn?	**Ectopic pregnancy in the horn**, which ruptures (often in the second trimester) with massive haemorrhage. Also haematometra and endometriosis. **Excise a functional horn.**
16	Obstetric complications of uterine anomalies?	Miscarriage, **preterm birth**, cervical insufficiency, **malpresentation (breech, transverse)**, FGR, retained placenta, PPH, higher CS rate, and rudimentary horn rupture.
17	Which fusion anomaly generally has the best reproductive outcome?	**Uterus didelphys** (each hemi-uterus is well vascularised). Metroplasty is not recommended.
18	Options for menstrual suppression while awaiting surgery for obstruction?	**Norethisterone 5 mg orally 2–3 times daily**, a **continuous COC**, or a **GnRH agonist** (specialist, e.g. goserelin 3.6 mg SC monthly with add-back). It is a bridge, not a cure.
19	Commonest cause of ambiguous genitalia in a 46,XX newborn, and its danger?	**Congenital adrenal hyperplasia** (21-hydroxylase deficiency). There is a risk of a **salt-wasting crisis** at 1–3 weeks. Check 17-OHP, electrolytes and glucose. Give IV hydrocortisone and saline.
20	Gonadectomy timing in CAIS vs Swyer syndrome?	**CAIS**: usually **after puberty** (natural breast development from aromatised testosterone; low tumour risk before adulthood), then oestrogen replacement. **Swyer (46,XY gonadal dysgenesis)**: **at diagnosis**, because of the high gonadoblastoma risk.
21	A pubertal girl with no periods presents in acute urinary retention. First steps?	**Urine pregnancy test**, **catheterise** gently and record the volume, analgesia, then **look at the introitus**: a bulging bluish membrane = imperforate hymen with haematocolpos. Confirm on ultrasound and scan the **kidneys**. Never needle-aspirate.
22	When should primary amenorrhoea be investigated?	No periods by about **15 years with normal breast development** (older texts say 16), or by **13 years with no breast development**; sooner if there is **cyclical pain, a pelvic mass or retention**.
23	Which examination findings separate MRKH from CAIS at the bedside?	Both: normal breasts, short blind vagina, no uterus. **MRKH**: normal pubic and axillary hair, ovaries on scan. **CAIS**: **scant pubic/axillary hair**, **inguinal masses (testes)**. Confirm with **testosterone** (female vs male range) and **karyotype** (46,XX vs 46,XY).
24	Which imaging is needed before surgery for a transverse vaginal septum, and why?	**MRI pelvis** to define the **level and thickness** of the septum and confirm a cervix is present (vs cervical agenesis). The first operation is the best chance; a thick or high septum needs an experienced surgeon, dilators and sometimes grafts. Suppress menses meanwhile.
25	Workup of a newborn with ambiguous genitalia who is vomiting at 2 weeks?	**Salt-wasting CAH** until proven otherwise: glucose, **electrolytes (low Na^+, high K^+)**, **17-OH progesterone** and karyotype; IV saline, glucose and **hydrocortisone 25 mg IV stat**; paediatric endocrinology/MDT. Do not assign sex at the bedside.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 14-year-old girl presents with acute urinary retention. She has never menstruated but has had monthly lower abdominal pain for 5 months. Breasts are Tanner 4. There is a tense, bluish bulging membrane at the introitus and a suprapubic mass. The pregnancy test is negative. After catheterisation, what is the MOST appropriate definitive management?

- A Needle aspiration of the mass through the membrane
- B Laparotomy to drain the haematometra
- C Cruciate (or elliptical) hymenotomy with drainage under anaesthesia
- D Start the combined pill and review in 3 months
- E Karyotype and wait for the result before any intervention

2. A 17-year-old with primary amenorrhoea has normal breast development and normal pubic and axillary hair. She has no cyclical pain. The vagina is a 2 cm blind pouch, and ultrasound shows no uterus but normal ovaries. What is the MOST likely karyotype and diagnosis?

- A 46,XY: complete androgen insensitivity syndrome
- B 45,X: Turner syndrome
- C 46,XY: Swyer syndrome
- D 46,XX: imperforate hymen
- E 46,XX: MRKH syndrome

3. A 16-year-old 'girl' with primary amenorrhoea has well-developed breasts, very scanty pubic and axillary hair, a short blind vagina and bilateral inguinal swellings. What investigation is MOST diagnostic?

- A Karyotype
- B Serum prolactin
- C Hysterosalpingogram
- D Pituitary MRI
- E Laparoscopy for ovarian biopsy

4. A 15-year-old has regular periods since age 13 but increasingly severe dysmenorrhoea. On examination there is a tense cystic mass bulging from the right lateral vaginal wall. Ultrasound shows two uterine horns with two cervices, and an absent right kidney. What is the BEST management?

- A Right hemi-hysterectomy
- B Continuous oral contraceptive pill indefinitely
- C Needle aspiration of the cystic mass
- D Resection of the obstructing vaginal septum to drain the hemivagina
- E Laparoscopic excision of the right ovary

5. A 28-year-old with three consecutive first-trimester miscarriages has an HSG showing two divergent uterine cavities. What is the MOST appropriate next investigation before deciding on surgery?

- A Repeat HSG
- B 3D transvaginal ultrasound or pelvic MRI to assess the external fundal contour
- C Karyotype of the woman only
- D Serum AMH
- E Endometrial biopsy

6. A woman with a known unicornuate uterus presents at 16 weeks' gestation with sudden severe abdominal pain, BP 70/40 and a pulse of 140. An earlier scan reported the pregnancy 'eccentrically placed'. What is the MOST likely diagnosis and immediate management?

A Placental abruption; deliver vaginally
B Red degeneration of a fibroid; give analgesia
C Threatened miscarriage; bed rest
D Ovarian torsion; laparoscopy tomorrow
E Ruptured rudimentary horn pregnancy; resuscitate and perform an emergency laparotomy

7. A 19-year-old with MRKH syndrome, recently married, is referred because her husband's family demands 'an operation to make her a woman'. She is distressed and unsure. What is the MOST appropriate initial approach?

A Immediate McIndoe vaginoplasty to save the marriage
B Tell the husband's family the diagnosis first
C Private, staged counselling with her, psychological support, and offering vaginal dilator therapy as first line when she is ready
D Refer for hysterectomy
E Advise that nothing can be done

8. Which anomaly results from failure of resorption of the midline wall after complete fusion of the Müllerian ducts?

A Septate uterus
B Bicornuate uterus
C Uterus didelphys
D Unicornuate uterus
E MRKH syndrome

9. A 15-year-old has cyclical pelvic pain and primary amenorrhoea. The hymen looks normal and is patent, but the vagina ends blindly at 3 cm. A cystic mass is felt above it on rectal examination, and ultrasound shows a uterus with haematometra. What is the NEXT step?

A Hymenotomy in the ward
B Needle aspiration through the vaginal vault
C Karyotype
D Pelvic MRI to define the level and thickness of a transverse septum, with menstrual suppression and referral to a specialist surgeon
E Hysterectomy

10. In a woman newly diagnosed with a unicornuate uterus, which additional investigation is MOST important?

A Thyroid function tests
B Renal ultrasound
C Echocardiogram
D Bone density scan
E Serum CA-125

11. A 30-year-old with a known bicornuate uterus is 16 weeks pregnant. Her previous pregnancy ended in a painless delivery at 22 weeks. What is the MOST appropriate antenatal plan?

A Hysteroscopic metroplasty now
B Termination of pregnancy
C Serial transvaginal cervical length scans, with cerclage and/or vaginal progesterone if the cervix shortens
D Routine care only; bicornuate uterus does not affect pregnancy
E Bed rest until 37 weeks

12. A 3-week-old baby with ambiguous genitalia (clitoromegaly, fused labia, no palpable gonads) presents with vomiting, lethargy, Na^+ 124 mmol/L and K^+ 7.0 mmol/L. What is the MOST appropriate immediate management?

A Assign male sex and arrange surgery
B Pelvic MRI before any treatment
C Oral rehydration solution and discharge
D Karyotype and wait for the result before treatment
E IV 0.9% saline, glucose if hypoglycaemic, and IV hydrocortisone after taking blood for 17-OH progesterone

13. A primigravida at 36 weeks has a persistent breech presentation. An early pregnancy scan noted a 'heart-shaped uterus with a deep external fundal cleft'. What is the MOST appropriate advice about ECV?

A A major uterine anomaly is a contraindication to ECV; discuss planned CS versus vaginal breech birth
B ECV is recommended and highly likely to succeed
C Perform ECV without tocolysis at 36 weeks
D Perform internal podalic version in labour
E Induce labour at 37 weeks

14. Which statement about the embryology of the female genital tract is CORRECT?

A The ovaries develop from the Müllerian ducts
B The whole vagina is derived from the Müllerian ducts
C AMH is required for development of the uterus
D The lower vagina and hymen derive from the urogenital sinus, and the upper vagina from the Müllerian ducts
E The Wolffian ducts form the fallopian tubes

15. A 22-year-old with a complete longitudinal vaginal septum and two cervices (didelphys) is in labour at 39 weeks, with a cephalic presentation at full dilatation. The septum is stretched tightly across the descending head. What is the BEST management?

A Immediate caesarean section
B Double-clamp, divide and ligate the septum, then allow vaginal delivery
C Vacuum delivery with the septum in place
D Wait for the septum to tear spontaneously
E Symphysiotomy

Answers & explanations

> **1. C** · Cruciate (or elliptical) hymenotomy with drainage under anaesthesia
>
> This is an **imperforate hymen with haematocolpos**. Treat by **hymenotomy** (cruciate or elliptical incision, excision of the flaps, suture of the mucosa to the ring) and drainage. **A** Needle aspiration drains incompletely and risks **pyocolpos**. **B** Laparotomy is unnecessary. **D** Suppression doesn't drain the collected blood. **E** Secondary sexual characteristics and the obvious membrane make a DSD unlikely, and a delay prolongs the obstruction.

2. E · 46,XX: MRKH syndrome

Normal breasts **and normal pubic hair**, no uterus, a blind vagina and **normal ovaries** mean **MRKH (46,XX)**. **A** CAIS has scant pubic hair and testes, not ovaries. **B** Turner syndrome causes absent puberty and a uterus is present. **C** Swyer syndrome has streak gonads, no breast development and a uterus. **D** An imperforate hymen causes cyclical pain and a uterus is present.

3. A · Karyotype

This is **CAIS**: breasts from aromatised testosterone, **absent androgen effect on hair**, and inguinal testes. The **karyotype (46,XY)** confirms it, with male-range testosterone. **B/D** are for hypothalamic or pituitary causes. **C** is impossible (no uterus) and unhelpful. **E** is invasive and unnecessary.

4. D · Resection of the obstructing vaginal septum to drain the hemivagina

This is **OHVIRA (Herlyn–Werner–Wunderlich)**: didelphys, an obstructed right hemivagina, and right renal agenesis. **Septum resection** drains the blocked side and preserves both uteri. **A** is unnecessary and removes reproductive potential. **B** only suppresses and does not drain. **C** risks infection and re-accumulation. **E** is irrelevant.

5. B · 3D transvaginal ultrasound or pelvic MRI to assess the external fundal contour

An HSG shows only the cavity, so it **cannot distinguish septate from bicornuate**. **3D US or MRI** shows the external fundal contour. A septate uterus may benefit from hysteroscopic resection; a bicornuate uterus would be **perforated** by it. **A** gives the same limitation. **C** Parental karyotypes are part of the recurrent-loss work-up but don't define the anomaly. **D/E** don't assess anatomy.

6. E · Ruptured rudimentary horn pregnancy; resuscitate and perform an emergency laparotomy

A pregnancy in a **non-communicating rudimentary horn** typically ruptures in the **second trimester** with massive haemoperitoneum. Resuscitate and do a **laparotomy with excision of the horn**. **A** Abruption at 16 weeks is unlikely and would not explain the horn history. **B/C/D** do not cause this degree of shock, and delay would be fatal.

7. C · Private, staged counselling with her, psychological support, and offering vaginal dilator therapy as first line when she is ready

MRKH care starts with **confidential, staged counselling and psychological support**. **Dilator therapy** is first line when **she** is motivated, and surgery is reserved for failure. **A** Surgery under coercion, without trying dilators, is inappropriate. **B** breaches confidentiality. **D** There is no uterus to remove. **E** is false: a functional vagina can be created, and motherhood is possible via IVF with a gestational carrier.

8. A · Septate uterus

Failure of **septal resorption** gives a **septate** uterus (normal external contour). **B/C** are lateral **fusion** failures (partial or complete). **D** is failure of one duct to develop. **E** is agenesis of both ducts.

9. D · Pelvic MRI to define the level and thickness of a transverse septum, with menstrual suppression and referral to a specialist surgeon

A patent hymen with a **short blind vagina and haematocolpos above it** suggests a **transverse vaginal septum**. **MRI** defines its level and thickness for **specialist excision with anastomosis**. Suppress menses meanwhile. **A** The hymen is not the problem. **B** risks pyocolpos. **C** A uterus with menses implies a 46,XX female; a karyotype is not the priority. **E** is inappropriate.

10. B · Renal ultrasound

About **40%** of women with a unicornuate uterus have **renal anomalies** (usually ipsilateral agenesis), because of the shared Wolffian and Müllerian development. **A/C/D/E** have no specific association.

11. C · Serial transvaginal cervical length scans, with cerclage and/or vaginal progesterone if the cervix shortens

A bicornuate uterus with a previous painless mid-trimester loss suggests **cervical insufficiency**. **Cervical surveillance** with cerclage and/or progesterone as indicated is appropriate. **A** Surgery is contraindicated in pregnancy, and hysteroscopic metroplasty is for a septate uterus. **B** is not indicated. **D** is false: preterm risk is increased. **E** Bed rest is not effective and increases VTE risk.

12. E · IV 0.9% saline, glucose if hypoglycaemic, and IV hydrocortisone after taking blood for 17-OH progesterone

This is a **salt-wasting crisis of CAH**. Treat immediately with **IV fluids, glucose and hydrocortisone** after drawing diagnostic bloods (17-OHP, electrolytes). **A** Sex assignment is never urgent or unilateral. **B/D** delay life-saving treatment. **C** is dangerous with severe hyperkalaemia and hyponatraemia.

13. A · A major uterine anomaly is a contraindication to ECV; discuss planned CS versus vaginal breech birth

A **bicornuate uterus** (a major uterine anomaly) is a **contraindication to ECV** (RCOG), and the breech is often due to the shape of the uterus. Counsel on the mode of delivery. **B/C** are inappropriate. **D** is only for a second twin. **E** Induction doesn't correct the breech.

14. D · The lower vagina and hymen derive from the urogenital sinus, and the upper vagina from the Müllerian ducts

The **upper vagina is Müllerian** and the **lower vagina and hymen are from the urogenital sinus** (sinovaginal bulbs and the vaginal plate). **A** The ovaries come from the genital ridge. **B** Only the upper part is Müllerian. **C** AMH causes Müllerian **regression**; its absence allows the uterus to form. **E** The tubes are Müllerian; Wolffian remnants are Gartner's duct cysts.

15. B · Double-clamp, divide and ligate the septum, then allow vaginal delivery

A **longitudinal septum** obstructing the second stage can be **divided between clamps and ligated**, allowing vaginal birth. **A** CS is unnecessary for a simple septum with a cephalic presentation at full dilatation. **C** Traction against the septum causes tears. **D** risks an uncontrolled tear and haemorrhage. **E** is obsolete and inappropriate.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: The 'pregnant' 14-year-old

A 14-year-old girl is brought to casualty at a regional referral hospital with inability to pass urine for 12 hours and a lower abdominal swelling. Her mother fears she is pregnant. She has never menstruated. For 6 months she has had lower abdominal pain for about 5 days each month.

1. What focused history and examination will you do?

> **History**:
>
> - Breast development and pubic hair.
> - The cyclical nature of the pain.
> - Bowel habit.
> - Sexual activity (confidentially).
> - Growth.
> - Past urinary problems.
>
> **Examination**:
>
> - Tanner staging.
> - Abdomen (a mass, its size, tenderness, a palpable bladder).
> - **Inspect the vulva gently**: look for a **bulging bluish membrane**.
> - Hernias.
> - Rectal exam if needed, for an anterior cystic mass.

2. She is Tanner 4. There is a tense blue bulge at the introitus. What are your immediate actions and investigations?

> - **Urethral catheter** to relieve the retention. Analgesia.
> - **Urine pregnancy test** (still essential).
> - **Ultrasound**: haematocolpos, the uterus, the ovaries and the **kidneys**.
> - FBC.
> - Explain the likely diagnosis kindly to her and her mother: **this is not pregnancy**, and it is easily treated.

3. Ultrasound shows a 12 cm haematocolpos with mild haematometra. What definitive treatment and what must you avoid?

> **Hymenotomy under anaesthesia**:
>
> - a cruciate or elliptical incision
> - drain the old tarry blood
> - excise the flaps
> - suture the vaginal mucosa to the hymenal ring with fine absorbable sutures
>
> **Avoid** needle aspiration (**pyocolpos**) and instrumenting the uterus.

4. What follow-up and counselling will you give?

> - Expect normal periods and future fertility.
> - Review in 4–6 weeks to check patency.
> - Watch for **infection** (fever, foul discharge).
> - Report any severe dysmenorrhoea later (endometriosis after a long obstruction).
> - Reassure her about normal sexual and reproductive function.

> **Take-away:** Primary amenorrhoea with cyclical pain and a pelvic mass in a pubertal girl is an outflow obstruction until proven otherwise. Look at the hymen before you think of pregnancy or a tumour, and never needle it.

Case 2: No periods, no pain

An 18-year-old university student attends the gynaecology clinic at Mengo because she has never had a period. She is otherwise well, is 162 cm tall, and plans to marry next year.

1. What key questions and examination findings will guide you?

> **Ask about**:
>
> - Breast development and pubic and axillary hair.
> - **Cyclical pain** (obstruction).
> - Weight, exercise, stress.
> - Sense of smell (Kallmann).
> - Headaches and galactorrhoea.
> - Family history.
> - **Sexual activity**.
>
> **Examine for**:
>
> - Tanner stage and **hair distribution**.
> - **Inguinal masses**.
> - Turner features.
> - External genitalia and **vaginal length** (a gentle swab).

2. She has Tanner 5 breasts, normal pubic hair, no inguinal masses, and a 2 cm blind vaginal dimple. What is your differential, and which tests do you order?

> **MRKH** is most likely (normal hair). The differentials are **CAIS** (hair usually scant), a transverse septum or cervical agenesis (but no cyclical pain), and constitutional delay (unlikely at 18).
>
> **Tests**:
>
> - Pregnancy test.
> - **Pelvic ultrasound or MRI** (uterus absent? ovaries present?).
> - **Karyotype**.
> - Testosterone, FSH, LH, oestradiol.
> - **Renal ultrasound**.
> - Spine X-ray and hearing test if indicated.

3. The results confirm MRKH (46,XX, absent uterus, normal ovaries, left pelvic kidney). How do you break the news?

> - A private room, with her own choice of support person.
> - Check what she knows, and give a warning shot.
> - Explain simply that the womb and upper vagina did not form before birth, but **her ovaries and hormones are normal, and she is fully a woman**.
> - **A vagina can be created** (dilators first).
> - **Genetic motherhood** is possible through IVF with her eggs and a gestational carrier, or adoption.
> - Offer psychology and peer support.
> - Staged follow-up visits.
> - Give her time and written information.

4. What is the plan for creating a vagina, and what about her pelvic kidney?

> **Vagina**:
>
> - **Vaginal dilator therapy** (Frank method) when she is ready: graduated dilators for about 20–30 min once or twice daily, with nurse or physiotherapy support.
> - Surgery (McIndoe, Vecchietti, Davydov) if it fails.
> - Ongoing dilation or intercourse maintains the result.
>
> **Kidney**:
>
> - Note the **pelvic kidney** for any future pelvic surgery.
> - Check BP and renal function periodically.

Take-away: Primary amenorrhoea with normal breasts and pubic hair but no uterus is MRKH. Karyotype, scan the kidneys, counsel with great sensitivity, and use dilators first.

Case 3: Recurrent losses and a breech

A 31-year-old para 1+3 attends ANC at 12 weeks. She has had three miscarriages at 8–10 weeks and one breech delivery by CS at 36 weeks, at which the surgeon wrote 'uterine anomaly?'. She has never had imaging of her uterus.

1. What anomaly is suspected, and how will you investigate now and after this pregnancy?

> A **septate** (or bicornuate) uterus is suspected: recurrent first-trimester losses, a breech, and the intra-operative finding.
>
> **Now**:
>
> - Early ultrasound to confirm an intrauterine pregnancy in the main cavity.
> - Check which side, and exclude a rudimentary horn.
> - Renal scan.
>
> **After pregnancy**: **3D ultrasound or MRI** to assess the external fundal contour; hysteroscopy (± laparoscopy) as needed.

2. How will you manage this pregnancy?

> - **Serial cervical length scans** from 16 weeks (cerclage or progesterone if short).
> - Growth scans.
> - Low threshold for review with bleeding or pain.
> - Check the presentation at 36 weeks: **ECV is relatively contraindicated** with a major anomaly.
> - Plan delivery (she also has a **previous CS**).
> - Be ready for retained placenta and PPH.

3. After delivery, 3D ultrasound shows a normal external fundal contour with a 2.5 cm internal septum. What do you offer?

> A **septate uterus**. For a woman with recurrent pregnancy loss, offer **hysteroscopic septum resection** after counselling:
>
> - the evidence of benefit is debated in unselected women
> - the risks are perforation and adhesions
> - rarely, uterine rupture in a later pregnancy
>
> Plan it for **at least 3 months postpartum**.

4. Why would the same operation be dangerous if the uterus were bicornuate?

> In a **bicornuate** uterus the 'septum' is actually **full-thickness myometrium** between two horns, with an external cleft. Hysteroscopic resection would **perforate** into the peritoneal cavity, with a risk of bowel or vascular injury. That is why you must define the external contour first.

> **Take-away:** Recurrent miscarriage plus malpresentation suggests a uterine anomaly. Define the external contour before any surgery: only a septum is resected.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.